

Empirical Formulas Between Outdoor Temperature And Humidity

1st Sensus Wijonarko
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
19w18r26@gmail.com

2nd Mahmudi
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
mahmudiku@yahoo.com

3rd Tatik Maftukhah
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
maftukahah@gmail.com

4th Dadang Rustandi
Instrumentation Development Unit
BRIN
Bandung, Indonesia
drustandi60@gmail.com

5th Purwowibowo
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
purwowibowo@gmail.com

6th Bambang Widiyatmoko
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
widiyatmoko@yahoo.com

7th Dwi Bayuwati
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
bayuwati@gmail.com

8th Bernadus H Sirenden
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
bernadushs@gmail.com

9th Jalu A Prakosa
Research Center for Physics
BRIN
Tangerang Selatan, Indonesia
jaluahmad@gmail.com

10th Asep Rahmat Hidayat
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
asep@gmail.com

11th Dwi Mandaris
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
dwi@gmail.com

12th Himma Firdaus
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
himma@gmail.com

13th Siddiq Wahyu Hidayat
Research Center for Testing Technology
BRIN
Tangerang Selatan, Indonesia
siddiq@gmail.com

Abstract— Outdoor temperature and humidity are vital variables for many aspects of life, such as determining the electrical and electronic component lifetime. However, an existing relationship between outdoor temperature and humidity in our ordinary mind is in exponential form. Hence, it isn't easy to imagine their correlation straightly. This study was devoted to obtaining a more straightforward format to connect outdoor temperature and humidity. More than fifty thousand latest data were collected using automatic measurements to both variables for 12 months at our Hydrometeorological Monitoring Station. This station location is on Research Center for Physics – BRIN (*Badan Riset Dan Inovasi Nasional*, National Research and Innovation Agency Republic of Indonesia). Data analysis used the piecewise linear regression method, a method that is usually used for diode analysis. This method was also suitable with the collected data, and the purpose of this research was to make a simple relation between temperature and humidity. The result shows that there were 12 linear equations where their constants were a little bit different as a function of the month. Applying one of these equations to the suitable case, we can predict humidity data lost if we know the temperature and vice versa. This study should be expanded in the future for data from four-season countries due to contrasting

climates between tropical countries and subtropical countries.

Keywords—*temperature, humidity, temperature and humidity correlation, piecewise linear regression method*

I. INTRODUCTION

Outdoor temperature and humidity are two of several hydrometeorological variables used as natural media to govern the existence of life. Initially, the earth's mean temperature -mainly due to the sun radiation - over all latitudes and all seasons is -23°C. Still, due to the natural greenhouse effect by water vapour, carbon dioxide, nitrous oxide, methane and ozone, the earth mean temperature becomes 15°C [1]. Fortunately, the outdoor temperature and humidity are not distributed uniformly around the world. All of these processes enable the existence of the world hydrological cycle and vegetation photosynthesis. The water cycle, food web, and oxygen exchange are the essential prerequisites for the existence of creatures.

The moderate temperature and humidity are comfortable and healthy. Temperature and relative humidity are the most important factors affecting human comfort [2]. The average temperature of 15°C is located in the temperature range of many plantations, animals, and human beings live. Thermal comfort is essential for people to live and work well in this world. The impact of

humidity on human responses was significant and increased with an increase in air temperature when the relative humidity was above 70% [3].

Extreme temperature and humidity are dangerous for life. Extreme temperature is harmful to human health because the human body will maintain its temperature to 37 ° C. Excessive humidity is also detrimental to human health because the humid atmosphere helps the growth of bacteria and germs and causes laziness and lethargy [4]. Heat stress caused by high - extremely hot and humid environment and valuated using moist enthalpy and moist entropy – and even extreme heat stress has become a frequent extreme weather event around the world, especially in megacities [5]. Heatstroke – due to overexposure of ambient temperature (AT) and relative humidity (RH) – threatens organ damage or death [6]. The number of daily new confirmed cases to be positively associated with both average temperature and relative humidity [7]. Temperature and humidity affect the ageing of equipment [8]. They can produce changes in the ambient conditions (temperature and humidity) of the air layers around them, which create an exciting insulation effect [9].

Temperature and humidity variables are also used for many necessities of life. They are used as controller variables to reduce energy consumption [10]. Changes in the temperature are used by a pair of temperature sensor to measure the airflow [11]. They are restrictors for the growth of plantations such as oyster mushrooms [12]. Controlling the temperature of ingredients plays a critical role in creating a delicious meal [13]. Atmospheric temperature and humidity are parameters of great interest in precision agriculture or air quality assessment in urban agglomerations [14]. Measured data of ambient temperature and relative humidity provide an alternative method to quantify the atmospheric turbidity and solar irradiance, important variables for the design and performance of solar power plants, soiling evaluation of solar collectors, energy-saving, and monitoring the global warming [15]. Measurement of humidity and temperature is an important task to be solved in environmental monitoring, control of technological processes, storage conditions for foodstuff, works of art, in greenhouses, as well as air control indoors [16]. Temperature and relative humidity are two of the most important variables in breath analysis [17]. Temperature and humidity are vital parameters in monitoring the health of patients in critical care [7] & [18].

The temperature has many relationships with other hydrometeorological variables, such as temperature and total solar radiation [19] and temperature and humidity. Naturally, the relation between temperature and humidity is exponential and depends on other variables. One of these relations is presented as:

$$T_0 = Ke^{\left(\frac{H_{th}}{H_0}\right)^2} \quad (1)$$

where T_0 and H_0 are the experimental values of the temperature and humidity, a Gaussian trend for humidity versus temperature [20]. Hence, it is difficult for us to

compute the relation values between temperature and humidity in our minds.

This study aimed to attain a more straightforward formula to relate outdoor temperature and humidity utilizing the linear regression method for both variables obtained from our Hydrometeorological Station, Research Centre for Physics – BRIN, during 12 months. The obtained equation can make us easier to predict the temperature data loss from humidity data and vice versa without measuring other hydrometeorological variables.

II. AN EXISTING TEMPERATURE AND HUMIDITY CORRELATION

Temperature is a measure of the degree of hotness or coolness of a body [1]. Hotness or coolness is a sensation of heat, while heat is one energy needed for our life.

On the earth, temperature primarily depends on the amount of solar radiation received by it. The position of the earth covered by the sun radiation is shifted over time. Therefore, there are three temperature zone, the torrid zone, temperate zone, frigid zone. The torrid zone is the region lying between the tropic of cancer and the tropic of Capricorn. Temperate zone is The region between the Tropic of Cancer ($\pm 23\frac{1}{2}^\circ$ N) and $\pm 66\frac{1}{2}^\circ$ N in the northern hemisphere and between the Tropic of Capricorn ($\pm 23\frac{1}{2}^\circ$ S) and $\pm 66\frac{1}{2}^\circ$ S in the southern hemisphere. The frigid zone is the region between $\pm 66\frac{1}{2}^\circ$ N and the north pole in the northern hemisphere and between $\pm 66\frac{1}{2}^\circ$ S and the south pole in the southern hemisphere.

Humidity is the amount of water vapor present in the atmosphere [1], [2], & [21]. It is usually specified as relative humidity (RH), namely

$$RH = \frac{e}{e_s} \times 100\% \quad (2)$$

where e_s is saturated vapor pressure (mbar)

$$e_s = 6.107 \times 10^{\frac{7.5T}{(237+T)}} \quad (3)$$

and T is the ambient temperature.

A psychrometer - hygrometer with dry and wet probes [2] - is a common instrument for determining the air relative humidity. This is a pair of wet and dry bulb thermometers. The wet-bulb thermometer is shrouded by a piece of ever wet muslin because it is connected to a wick dipped in a small water bottle. The dry bulb thermometer is bare. The difference between the temperature ($^\circ\text{C}$) readings ($T - T_w$) of the dry-bulb thermometer and the wet bulb thermometer is a measure of the actual vapour pressure presented as

$$e = e_s - 0.00066 P(T - T_w)(1 + 0.00115T_w) \quad (4)$$

where P = the pressure of the air (mb)

There is a significant difference between outdoor and indoor impact for temperature and humidity. Humidity depends on outdoor temperature [22], but temperature and humidity in an indoor environment can be controlled independently [23]. Organisms respond more to the local microenvironmental than large-scale conditions [24]. The discussion is limited to outdoor conditions only.

Most research on humidity sensors does not pay attention to the relationship between humidity and temperature [22]. That is why this paper presents the psychrometer theory. This sensor shows a clear relationship between both variables. Other sensors - such as membrane that permeable to water vapour and impermeable to liquid water [25], capillary hygrometer [2], and interferometer [26] and [27], SHT21 temperature and humidity sensors [28], Fabry-Perot interferometer (FPI) [29], and piezoresistive humidity sensor [30] - do not show this relationship clearly.

From (3) and (4), it can be attained that

$$RH = \frac{6.107 \times 10^{\frac{7.5T}{237+T}} - 0.00066 P(T-T_w)(1+0.00115T_w)}{6.107 \times 10^{\frac{7.5T}{237+T}}} \quad (5)$$

III. METHODOLOGY

III.1. Data Collection

Data were collected from our Hydrometeorological Station, Research Center for Physics – BRIN (Fig. 1) during the last 12 months or two-semester. Data were sent in a wireless way from this station to our computer that works especially as a data saver.



Fig. 1. Hydrometeorological equipment

III.2. Data Analysis

Data gathered from the data collection step were analyzed using Piecewise Linear Regression Model. The cutting side was based on the month. The collected data were classified in 12 months.

The linear regression model was analyzed using the linear analysis method. Regression analysis is a collection of statistical tools utilized to model and explore relationships between variables related in a nondeterministic manner [31]. The relationship between temperature and humidity in this study was classified as a nondeterministic relationship because this is a simplification for an exponential relationship that is also dependent on other variables such as pressure.

Regression analysis is valid if two prerequisites are fulfilled [31]. First, there are cause-and-effect relationships. Second, it works well only for values of the regressor variable within the range of the original data. Therefore, it is not necessarily valid for extrapolation purposes.

The regression analysis in this study was simple linear regression. This kind of regression has only a single

independent variable (regressor variable, predictor variable, x variable) [32] and a single dependent variable (response variable, Y variable). It is pretending that the true relationship between humidity (y) and temperature (x) is a straight line. Hence, the expected value or the mean of y for each value of x is

$$E(y|x) = \alpha x + \beta \quad (6)$$

where the slope α and the intercept β are unknown regression coefficients.

It is assumed that each observation, y, can be described by the model

$$y = \alpha x + \beta + \epsilon \quad (7)$$

where ϵ is a random error with mean zero and (unknown) variance σ^2 .

The estimation of α and β , namely $\hat{\alpha}$ and $\hat{\beta}$, should produce a line that is the best fit to the data. This is carried out by minimizing the sum of the squares of the vertical deviations (using the method of least squares) to produce

$$\hat{\alpha} = \frac{S_{xx}}{S_{yy}} \quad (8)$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n} \quad (9)$$

$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n} \quad (10)$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad (11)$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad (12)$$

$$\hat{\beta} = \bar{y} - \hat{\alpha} \bar{x} \quad (13)$$

The fitted or estimated regression line is

$$\hat{y} = \hat{\alpha} x + \hat{\beta} \quad (14)$$

$$y_i - \hat{y}_i = e_i \quad (15)$$

where e_i is the residual.

This study also used the piecewise regression linear method. In this case, the relationship of temperature and humidity were monthly, not yearly.

III.3. Data Presentation

There were thousands of data collected during 12 months. These data were not presented in a table but in a condensed figure. This figure was split monthly, so the data were presented as 12 in one figure.

IV. RESULTS AND DISCUSSION

IV.1. Results

The collected data (Fig. 2) shows that there was a uniformity among monthly data. However, the temperature data and humidity data opposed each other. When the temperature was high, the humidity was low, and vice versa. Briefly, both data were upset down.

IV.2. Discussion

Fig. 2 was analysed using the simple linear regression method as described in the methodology. TABLE 1 has uniform information because the data are uniform, as

presented in Fig. 2. Outdoor humidity (y) had a relation with outdoor temperature (x) in the form of $y = -\alpha x + \beta$. The minus sign means that the humidity and temperature were opposite each other. The lower the outdoor temperature, the higher the outdoor humidity.

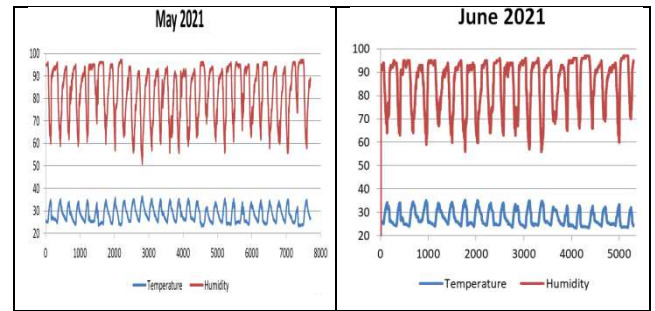
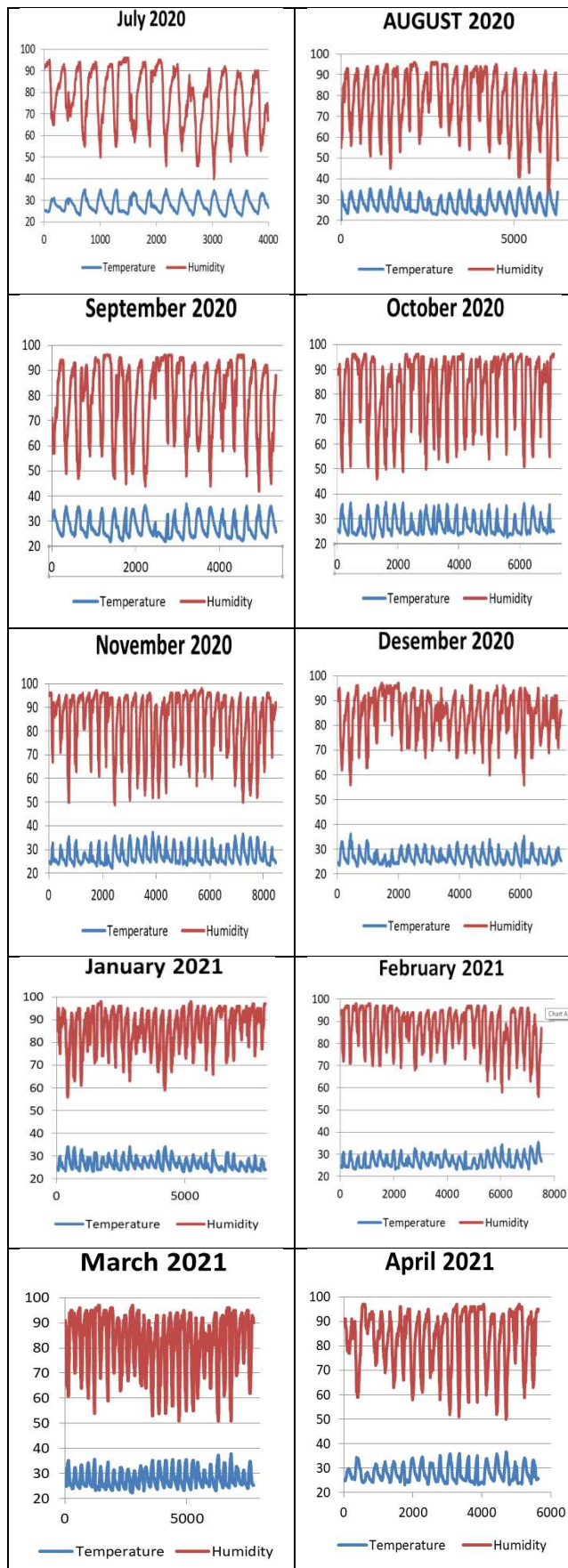


Fig. 2. The result

TABLE 1. FORMULAS BETWEEN HUMIDITY AND TEMPERATURE

Month	Formula	Month	Formula
1	$y = -3.3092x + 174.54$	7	$y = -3.7786x + 183.50$
2	$y = -3.3896x + 178.10$	8	$y = -3.6582x + 180.87$
3	$y = -3.2919x + 174.37$	9	$y = -3.6064x + 179.47$
4	$y = -3.4036x + 177.79$	10	$y = -3.6120x + 181.59$
5	$y = -3.4683x + 181.82$	11	$y = -3.4974x + 180.56$
6	$y = -3.1750x + 172.98$	12	$y = -3.1838x + 170.38$

At a glance, the simple linear regression $y = -\alpha x + \beta$ in (6) or Table 1 is not in line with (4). The temperature in that regression affect humidity. Hence, the temperature is the independent variable, while humidity is the dependent variable. On the other hand, there is a correlation between humidity and temperature in (4). Temperature affects humidity, and humidity affects temperature. Both equations will meet each other from the following descriptions.

There are some reasons why the gradient is minus? This finding is caused by two in one process. First, outdoor evaporation increases if the outdoor temperature is high [1] & [33]. The air temperature increases the water surface temperature, the vapour pressure gradient, the air capacity to absorb water vapour, and finally, the evaporation. Second, if the humidity of the air is getting higher, then evaporation will be getting lower [1] & [34] because water vapour during the process of evaporation moves from the higher vapour content point to the point of lower vapour content point.

It looks that α has a relation with the sun path above the earth. On June 21, the sun position is above The Tropic of Cancer (or the Northern Tropic, the sun position at the northernmost latitude 23.43°N , northern solstice) in the northern hemisphere where the sun above this region is directly overhead. Then the sun position shifts to the south and reaches southward equinox (autumnal equinox) on September 23. Next, the sun position is above The Tropic of Capricorn (the Southern Tropic), namely the southernmost latitude (23.43°S , southern solstice) on December 21. Then the sun reverses its direction to the north. the sunshine reaches the northward equinox (vernal equinox) on March 21 and continues its direction until it reaches the northern solstice. The whole process like this takes place once a year.

Fig. 3 shows that there were two question marks, namely the data in March and September. Why was the value of α in March not in between February and April, and the value of α in September was not between August and October? March and September are the time when the sun path is the above equator. The temperature in this place is high, so the evaporation is also high. On the contrary, evaporation decreases the strength of temperature (α).

Alpha

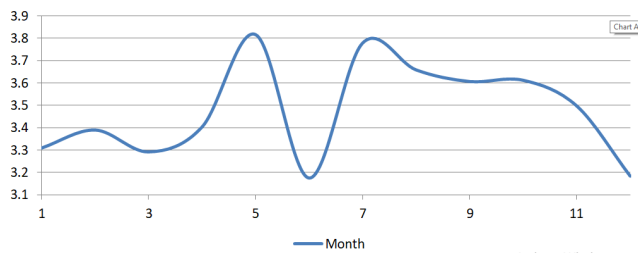


Fig. 3. The α peak changes

The β (Fig. 4) changes had a similar pattern as the α changes. This finding shows that the negative effect (α) and positive effect (β) to the humidity walked together. Briefly, the dominant factor for the humidity was temperature. Sun path had no significant impact.

Beta

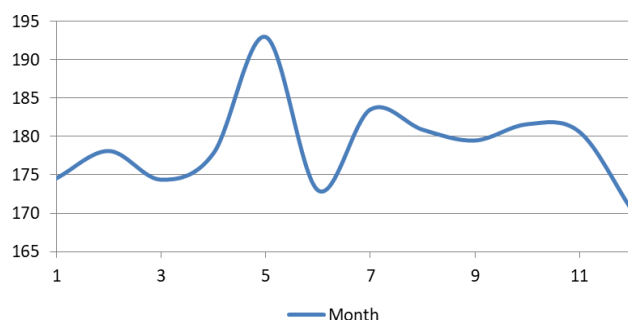


Fig. 4. The β peak changes

Based on the above explanations, it can be inferred that $y = -\alpha x + \beta$ is similar to (4). Although correlation, the effect of temperature to humidity is more dominant than the effect of humidity to temperature. Therefore (4) can be simplified as $y = -\alpha x + \beta$. The influence of humidity on outdoor temperature [35] in the simple linear regression was shown on the temporal and spatial aspects. The values of α and β vary in both time and space. These effects were clear from the piecewise analysis based on the month. Hence, there are no common values for α and β in this case. Every hydrometeorological station has its α and β best values to predict the humidity data loss from temperature, and estimate the temperature data loss from humidity.

An application example of this finding might be used for web-based rain gauge calibrators (WBRGCs). Every WBRGC whether in version 1 [36], version 2 [37] & [38], or higher version [39] has temperature and humidity instruments in it. The WBRGC is usually used in situ. Hence, before finishing the task, the WBRGC should give complete data. In case the temperature or humidity instrument does not perform properly, this finding can give a solution.

V. CONCLUSION

The theoretical correlation between temperature and humidity can be simplified using an empirical equation as $y = -\alpha x + \beta$, where y is humidity, x is temperature, and α and β are regression coefficients. This equation might be used particularly to estimate the humidity data loss from temperature, and vice versa. The effect of humidity to temperature is manifested by the influence of time and space to the values of α and β . Hence, the α and β values obtained in this study are climate-specific [40]. They might not be exactly the same if applied at other stations because the correlation between humidity and temperature is a function of region and season [41].

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REFERENCES

- [1] P. R. Rakhecha and V. P. Singh, *Applied Hydrometeorology*. New Delhi: Springer, 2009.
- [2] I. K. Al-Ataby and A. I. Altmimi, "Testing the Relationship between Air Temperature and Relative Humidity by Using T-Test for Some Selected Stations in Iraq," *Al-Mustansiriyah J. Sci.*, vol. 32, no. 2, pp. 1–7, 2021.
- [3] L. Jin, Y. Zhang, and Z. Zhang, "Human Responses to High Humidity in Elevated Temperatures for People in Hot-Humid Climates," *Build. Environ.*, 2017.
- [4] A. Laukkanen and J. Vinha, "Temperature and Relative Humidity Measurements and Data Analysis of Five Crawl Spaces," *Energy Procedia*, vol. 132, pp. 711–716, 2017.
- [5] S. Yang and E. Al., "Responses of Heat Stress to Temperature and Humidity Changes Due to Anthropogenic Heating and Urban Expansion in South and North China," *Front. Earth Sci.*, vol. 9, pp. 1–17, 2021.
- [6] K. Miyamoto and E. Al., "A Novel Mouse Model of Heatstroke Accounting for Ambient Temperature and Relative Humidity," *J. Intensive Care*, vol. 35, no. 9, pp. 1–11, 2021.
- [7] C. He and E. Al., "Optical fibre Sensor for Simultaneous Temperature and Relative Humidity Measurement: Towards Absolute Humidity Evaluation," *Sensors Actuators B. Chem.*, 2021.
- [8] C. J. Ryu and E. Al., "Evaluation of Temperature and Humidity of a Thermo-Hygrostat of PET/CT Equipment using a Temperature and Humidity Sensor(BME 280)," *J. Korean Soc. Radiol.*, vol. 14, no. 1, pp. 15–22, 2020.
- [9] P. Abdo, B. P. Huynh, and V. Avakian, "Effect of Green Wall Modules on Air Temperature and Humidity," in *Proceedings of the ASME 2018*, 2018, pp. 1–6.
- [10] S. Ye and E. Al., "Construction and Application of Temperature and Humidity Precision Control Model Based on Differentiation Theory," in *2021 International Conference on Power Grid System and Green Energy (PGSGE 2021)*, 2021, pp. 1–4.
- [11] G. V. Shevchenko, "Air Flow Sensor Based on Environmental Sensor BME280," in *21st International Conference On Micro/Nanotechnologies And Electron Devices EDM 2020*, 2020, pp. 432–435.
- [12] W. Adhiwibowo, A. F. Daru, and A. M. Hirzan, "Temperature and Humidity Monitoring Using DHT22 Sensor and Cayenne API," *Transformatika*, vol. 17, no. 2, pp. 209–214, 2020.
- [13] W. T. Sung, J. H. Chen, and C. L. Hsiao, "Data Fusion for PT100 Temperature Sensing System Heating Control Model Measurement," *Measurement*, vol. 52, pp. 94–101, 2014.
- [14] A. Zarnescu, "Study of the Waterproof Shelter

- Colour Influence on the Atmospheric Temperature and Humidity Measurements for an Internet of Things Application,” in *2018 IEEE 24th International Symposium for Design and Technology in Electronic Packaging (SIITME)*, 2018, pp. 217–220.
- [15] O. Behara, D. Sbarbaro, A. Marzoc, and L. Moran, “A simplified methodology to estimate solar irradiance and atmospheric turbidity from ambient temperature and relative humidity,” *Renew. Sustain. Energy Rev.*, vol. 116, pp. 1–6, 2019.
- [16] N. Nedev, Z. Nenova, and S. Ivanov, “Virtual Instruments for Humidity and Temperature Measurements,” 2013.
- [17] E. Mansour and E. Al., “Measurement of Temperature and Relative Humidity in Exhaled Breath,” *Sensors Actuators B. Chem.*, 2019.
- [18] J. Sun *et al.*, “Global evaluation of terrestrial near-surface air temperature and specific humidity retrievals from the Atmospheric Infrared Sounder (AIRS),” *Remote Sens. Environ.*, vol. 252, pp. 1–12, 2021.
- [19] P. E. Thornton and S. W. Running, “An Improved Algorithm for Estimating Incident Daily Solar Radiation from Measurements of Temperature, Humidity, and Precipitation,” *Agric. For. Meteorol.*, vol. 93, pp. 211–228, 1999.
- [20] S. Z. Ilyas, S. M. Nasiir, and Y. Kaplan, “Empirical relationship for humidity and temperature at Quetta, Pakistan,” *Math. Comput. Model.*, vol. 50, pp. 1495–1497, 2009.
- [21] C. Collier, *Hydrometeorology*. Chichester: John Wiley & Sons, Ltd, 2016.
- [22] H. M. Z. Yousaf and E. Al., “Highly Sensitive Wide Range Linear Integrated Temperature Compensated Humidity Sensors Fabricated using Electrohydrodynamic Printing and Electrospray Deposition,” *Sensors Actuators B Chem.*, vol. 308, pp. 1–11, 2020.
- [23] L. Yang and E. Al., “Independent Control of Temperature and Humidity in Air Conditioners by Using Fuzzy Sliding Mode Approach,” *Hindawi Complex.*, vol. 2020, pp. 1–12.
- [24] J. G. Mickley and E. Al., “Measuring Microenvironments for Global Change: DIY Environmental Microcontroller Units (EMUs),” *Methods Ecol. Evol.*, vol. 10, pp. 578–584., 2019.
- [25] R. Kellerman, “Method and Apparatus for Measuring Relative Humidity of a Mixture,” 2006.
- [26] R. J. Tong and E. Al., “Simultaneous Measurement of Temperature and Relative Humidity by Compact Mach-Zehnder Interferometer and Fabry-Perot Interferometer,” *Measurement*, vol. 155, pp. 1–6, 2020.
- [27] J. Zhang and E. Al., “Research on Simultaneous Temperature and Relative Humidity Measurement Based on Tapered PCF Mach-Zehnder Interferometer,” *Opt. Fiber Technol.*, vol. 61, pp. 1–7, 2021.
- [28] Q. Zhang, T. Guo, and A. Bao, “Design of Real-time Temperature and Humidity Measurement System Based on SHT21 Sensor,” in *2010 International Conference on Computer Application and System Modeling (ICCASM 2010)*, 2010, pp. V14-32–V14-35.
- [29] Y. He and E. Al., “Disriminative Measurement for Temperature and Humidity using Hollow_core Fabry-Perot Interferometer,” *Opt. Fiber Technol.*, vol. 53, pp. 1–6, 2019.
- [30] T. Waber and E. Al., “Fabrication and Characterization of a Piezoresistive Humidity Sensor with a Stress-free Package,” *J. Sens. Sens. Syst.*, vol. 3, pp. 167–175, 2014.
- [31] Anon, “Simple Linear Regression and Correlation,” pp. 401–512.
- [32] W. Navidi, *Statistics for Engineers and Scientists*. New York: McGraw-Hill Education, 2020.
- [33] J. D. Valiantzas, “Temperature-and humidity-based simplified Penman’s ET0 formulae. Comparisons with temperature-based Hargreaves-Samani and other methodologies,” *Agric. Water Manag.*, vol. 208, pp. 326–334, 2018.
- [34] T. G. Pypker, B. J. Bond, T. E. Link, D. Marks, and M. H. Unsworth, “The Importance of Canopy Structure in Controlling the Interception Loss of Rainfall: Examples from a Young and an Old-growth Douglas-fir Forest,” *Agric. For. Meteorol.*, vol. 130, no. 1–2, pp. 113–129, 2005.
- [35] K. Sene, *Hydrometeorology Forecasting and Applications*, 2nd ed. Heidelberg: Springer, 2016.
- [36] S. Wijonarko, T. Maftukhah, D. Rustandi, B. Sirinden, and N. T. E. Damayanti, “Web Based Rain Gauge Calibrator,” *J. Instrumentasi*, vol. 43, no. 1, pp. 25–42, 2019.
- [37] S. Wijonarko *et al.*, “The bias of web based rain gauge calibrator due to bubbles,” *J. Instrumentasi*, vol. 44, no. 1, pp. 1–12, 2020.
- [38] S. Wijonarko *et al.*, “The Second Generation of Web Based Rain Gauge Calibrator,” *J. Instrumentasi*, vol. 44, no. 2, pp. 113–121, 2020.
- [39] T. Maftukhah and Wijonarko. S., “A Data Processing Program For Web Based Rain Gauge Calibrators,” *J. Instrumentasi*, vol. 43, no. 1, pp. 53–64, 2019.
- [40] P. S. Mann, *Introductory Statistics*, 7th ed. Hoboken: John Wiley & Sons, Inc., 2017.
- [41] J. L. Nguyen and D. V. Dockery, “Daily indoor-to-outdoor temperature and humidity relationships: a sample across seasons and diverse climatic regions,” *Int J Biometeorol*, vol. 60, pp. 221–229, 2016.